Region-Calibrated Personas Do Not Significantly Improve Fit to CPT-Imputed Country-Level Choice Distributions: A Preregistered Null Result

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Abstract

Language models are increasingly used as stand-ins for human participants in behavioral research, yet they are systematically *more* rational than the people they simulate. One proposed fix infuses calibrated human-like irrationality at inference time; the open question is whether that calibration should be region-specific, given real cross-cultural variation in risk preferences. We preregistered exactly that test: do personas calibrated to *region-specific* measured preference distributions reproduce country-level choice heterogeneity better than a generic-human calibration? Running Qwen3-1.7B on CPU, we simulated individual respondents for ten countries spanning the empirical loss-aversion distribution, conditioning only on distribution-sampled behavioral profiles—no country label, ethnicity term, or trait inventory ever appears in a prompt—and compared simulated choice distributions against prospect-theory-imputed choice distributions (deterministic predictions from reconstructed within-country parameter distributions, not observed human choices) on six held-out willingness tasks. The preregistered primary test (paired permutation over countries, $\alpha = 0.05$) **rejects the hypothesis**: region-conditioned simulations average Wasserstein-1 distance 0.304 versus 0.285 for generic calibration (difference −0.019; one-sided $p = 0.631$), while the uncalibrated arm (0.294) matches about as well as either calibrated arm. This null concerns one small model under fixed protocols—not language models in general—and we discuss what it does and does not rule out for pluralistic alignment.

Keywords: cultural calibration; language models; behavioral simulation; null results; preregistration; prospect theory

Introduction

Language models are being recruited as simulated human subjects: they answer surveys, play economic games, and stand in for participants in market research (Argyle et al., 2023; Aher et al., 2023; Park et al., 2023). The practice rests on an untested premise: that a model's responses reproduce the decision patterns of the humans it replaces. The premise fails, and it fails in a specific, replicable direction. Across cognitive-reflection batteries, lottery choices, and repeated games, models deviate from humans toward *greater* rationality: they are too accurate (Aher et al., 2023), too close to expected-value maximization (Liu et al., 2025), and too consistent to be human (Macmillan-Scott & Musolesi, 2024). Instruction tuning can erase human-like biases that smaller base models display (Hagendorff et al., 2023). Call this the *rationality gap*.

If the gap is the disease, the natural treatment is deliberate: install calibrated irrationality by prompting or steering the model until its measured biases match human effect sizes. Early attempts show persona conditioning can elicit prospect-theory signatures (Dubedy, 2026) and motivated reasoning in models—yet none calibrates the installed irrationality against measured human distributions. That generic-calibration problem is taken up in a companion paper. Our starting observation: even a perfectly calibrated *generic* human is the wrong target, because human irrationality is not a single profile. Risk preferences vary substantially across countries (Rieger et al., 2015), loss aversion varies with culture (Wang et al., 2017), and the WEIRD-psychology literature shows that Western convenience samples are outliers on exactly the traits models default to (Henrich et al., 2010). A simulator calibrated to a single human profile flattens this heterogeneity.

The question we ask is narrower, and falsifiable: does calibrating irrationality levers *region-specifically*—to measured cross-country preference distributions—reproduce cross-country heterogeneity in economic choice better than a generic-human calibration? The ground truth is the INTRA survey (International Test on Risk Attitudes), which elicited risk preferences from roughly 7,000 respondents in 53 countries (Rieger et al., 2015), with per-country cumulative-prospect-theory parameters estimated in follow-up work (Rieger et al., 2017). To be precise about the sense in which this is "ground truth": the choice distributions we fit are not observed human choices but deterministic CPT predictions computed from *reconstructed* within-country parameter distributions (within-country SDs recovered from published SEs under an assumed $n = 150$, winsorized). What we measure is therefore how well qualitative personas track a parametric imputation of human heterogeneity—twice removed from actual choices—not direct human-likeness. The full design—countries, levers, tasks, metrics, tests, and decision rules—was preregistered before any model ran, and we report the preregistered test exactly as specified, null outcome included.

Three things come out of this work. First, a method for distribution-grounded regional calibration that satisfies a hard anti-essentialism constraint: levers are calibrated to measured *distributions* with within-country variance, prompts contain individual behavioral profiles rather than country labels, and culture is treated as situated and contextual (Oyserman & Lee, 2008) rather than as fixed national character. Second, a preregistered empirical test—ten countries spanning the measured loss-aversion distribution, region-conditioned versus generic-human versus uncalibrated simulation on held-out

willingness tasks, fit against prospect-theory-imputed choice distributions—that returns a null. Third, a constraint on the pluralistic-alignment agenda. Region-specific calibration, as implemented here, does not buy country-level fit. Along the way, a cautionary observation on response format: the Yes/No willingness format flips the usual rationality-gap pattern relative to the prospect-theory prediction. We report that flip as format sensitivity—not as a challenge to the human-data finding—and it limits what "human-like" can mean as a calibration target.

Related work

The rationality gap between LLMs and humans

Excess rationality is the most reliable fidelity failure in LLM-as-human simulation. Aher et al. (2023) named it *hyper-accuracy distortion*: aligned models are too factually correct to reproduce human response distributions. The direct decision-making statement comes from Liu et al. (2025): against the 13,000-choice human dataset choices13k, GPT-4o, Claude 3 Opus, and Llama-3 models deviate from humans toward expected-value theory, both when simulating and when predicting choices. Alignment itself is partly responsible—Hagendorff et al. (2023) show mid-scale GPT-3 gave the intuitive wrong answer on cognitive-reflection items about 80% of the time (humans about 55%), while instruction-tuned ChatGPT scored 59–96% correct. The settled picture: models are *differently* biased rather than uniformly biased (Pagliaro, 2026; Suri et al., 2024), and bias is a prompt–model interaction property, not a stable trait.

Attempts to make models irrational

Prompting can nudge models toward human-like bias, but without calibration. LLM "irrationality" is inconsistent across repeats and often non-human in form (Macmillan-Scott & Musolesi, 2024). Persona assignment does elicit motivated reasoning (Dash et al., 2025) and prospect-theory signatures—Dubedy (2026) reports GPT-4.1 with income personas reproducing the fourfold pattern unprompted. The closest concurrent work treats LLMs as calibrated measurement instruments for behavioral parameters: Yee & Koh (2026) induce large, stable shifts in loss aversion, herding, extrapolation, and anchoring toward human benchmark magnitudes across 24,000 agent–scenario pairs via profile-based calibration. Their calibration targets generic human benchmarks, however, not region-specific distributions, and they do not test whether calibration reproduces cross-country heterogeneity. We leave the generic-calibration problem to a companion paper and ask instead whether calibration should be regional.

Cross-cultural variation in economic preferences

Regional variation in irrationality is among the best-established facts in behavioral economics. Across 15 small-scale societies, Henrich et al. (2001) document dramatic cross-society variation in ultimatum, dictator, and public-goods play; Henrich et al. (2010) argue Western samples are frequent outliers. On risk, Rieger et al. (2015) surveyed 53 countries and found risk aversion in gains and risk seeking in losses everywhere—but at degrees that differ significantly across countries. Per-country cumulative-prospect-theory parameters estimated by Wang et al. (2017) show loss aversion varying systematically with culture. The largest validation is the Global Preference Survey: experimentally validated preference measures for 80,000 individuals in 76 countries, with substantial cross-country heterogeneity—and larger within-country heterogeneity (Falk et al., 2018). That literature gives us our design rule directly: calibrate to distributions, never to country means (Falk et al., 2018).

Simulating regional humans, and the stereotype trap

Country labels do shift model responses toward the target country (Durmus et al., 2023; Tao et al., 2024)—but they invite exactly the failure this paper is designed to avoid: Durmus et al. (2023) show country-perspective prompting can *amplify* cultural stereotypes, and Cheng et al. (2023a) and Cheng et al. (2023b) document stereotyped portrayals and caricature in persona-conditioned generations. The constructive alternatives are all distribution- or retrieval-grounded. Abels et al. (2026) build "calibrated value personas" from *individual-level* survey responses; Seo et al. (2025) retrieve per-individual value summaries instead of relying on model priors. We combine both ideas: individual-level profiles sampled from measured country distributions, and—critically—no country label anywhere in the prompt, so the model cannot fall back on training-data national associations. Our theoretical guardrail is Oyserman & Lee (2008): cultural influence is situated and moment-to-moment rather than a static national trait, so each prompt describes one respondent's measured survey answers, never a national character.

Method

We wrote the full preregistration—countries, levers, tasks, metrics, tests, decision rules, anti-essentialism rules, and two amendments—before any model ran. What follows summarizes the design *as run*, amendments included.

Ground truth: the INTRA survey

The ideal ground truth would be the Global Preference Survey (Falk et al., 2018), but its microdata require a data-request form and forbid redistribution, so we excluded it per the preregistration and fell back to the pre-specified alternative: the INTRA survey (Rieger et al., 2015), which elicited willingness-to-pay for gain lotteries and willingness-to-accept for loss lotteries from about 7,000 economics students in 53 countries, plus the per-country cumulative-prospect-theory parameters estimated from it (Rieger et al., 2017)—loss aversion λ, curvature ϕ, probability-weighting δ. Numerical values are the published per-country estimates as tabulated in Xie et al. (2018) (open access).

The INTRA publications report per-country standard errors rather than individual-level data, so we reconstruct within-country distributions parametrically: each individual's $(\lambda_i, \phi_i, \delta_i)$ is drawn from a truncated normal centered on the published country estimates, with SD = SE $\times \sqrt{n}$, $n = 150$ (median INTRA per-country sample size), winsorized to published ranges and λ truncated at 0.02 (Iceland's published $\lambda = -0.112$ becomes small-positive draws; disclosed). Absolute fit levels depend on this reconstruction, but the region-versus-generic comparison does not—both arms draw from the same reconstructed family—and sensitivity analyses scaling all within-SDs by 0.5 and 1.5 leave the ordering unchanged.

Country selection

We selected ten countries stratified across the empirical loss-aversion distribution (λ is the prospect-theory parameter shown to vary most with culture, Wang et al. (2017)): Iceland (−0.11), Greece (0.33), Brazil (0.60), Hungary (0.93), Pakistan (1.20), Australia (1.65), Germany (1.90), Japan (1.93), Denmark (2.21), Italy (3.35). Nothing but these published numbers drove the selection.

Elicitation and calibration levers (Amendment 2)

Pilot diagnostics forced an elicitation redesign before any main-run generation (preregistered Amendment 2). The chat template produced degenerate label bias—near-deterministic single-label outputs regardless of profile—which raw completion removes. Binary A/B choices proved content-insensitive to numerical certainty-equivalent profiles: first-token choice probabilities came out near-identical for Italy-like versus Hungary-like profiles. The final elicitation is therefore **raw completion** (no chat template) with **qualitative 3-level personas** and **Yes/No willingness questions**: e.g. "Would you pay $45 for a 50/50 chance to win $100? Answer:" with the first Yes/No token parsed.

Each simulated individual draws $(\lambda_i, \phi_i, \delta_i)$ either from their country's reconstructed distribution (**region** arm) or from the pooled multi-country mixture (**generic** arm); the draw is then mapped to a qualitative persona via absolute tertile cutoffs from the pooled reconstructed distribution. That mapping yields three risk sentences (from curvature ϕ, used on gain tasks) and three loss sentences (from loss aversion λ, used on loss tasks), e.g. "You are very cautious with money. You strongly prefer guaranteed amounts over any gamble." The **uncalibrated** arm sees the question with no persona. No prompt in any arm contains a country name, an ethnicity term, or a trait inventory—a scripted audit verifies this over all prompt templates and all 1,392 generations (0 hits).

Decision tasks

We held out six Yes/No willingness tasks with prices fixed across individuals, spanning the human certainty-equivalent range: pay $25, $45, or $65 for a 50/50 chance to win $100 (gain); accept a 50/50 chance of losing $100 if paid $40, $90, or $140 (loss). Each simulated individual's "real" counterpart is the deterministic choice implied by the $(\lambda_i, \phi_i, \delta_i)$ drawn for that individual under cumulative prospect theory (Kahneman & Tversky, 1979; Tversky & Kahneman, 1992): yes on a gain task iff price $\leq 100 \cdot 0.5^{\phi_i}$; yes on a loss task iff price $\geq 100 \cdot (0.5\lambda_i)^{\phi_i}$. We use the Tversky–Kahneman one-parameter weighting form because the exact form in Rieger et al. (2017) is not public (disclosed assumption).

Model and scale

We ran Qwen3-1.7B (Q4_K_M GGUF) with CPU-only inference via llama.cpp (temperature 0.7, top_p 0.8, top_k 20, max_tokens 24, n_ctx 1024—reduced from the preregistered 64 and 2048 under Amendment 2's elicitation redesign, which replaced the chat template and its `/no_think` suffix with raw completion). The preregistered scale after Amendment (N_INDIV 16 → 8, to bound CPU time) is 8 individuals × 10 countries × 2 conditioned arms × 6 tasks = 960 main generations, plus 48 uncalibrated baseline generations and 384 robustness generations (prompt paraphrase, seed variant, temperature 0.3, and a loss-aversion-only ablation, on Germany and Italy): 1,392 generations in total. Seeds are deterministic (master seed 20261008); A/B option order does not apply in the Yes/No format.

Metrics and preregistered tests

Our **primary** metric is the per-country Wasserstein-1 distance between the simulated per-individual risky-choice distribution and the prospect-theory-imputed distribution, averaged over the ten countries; the preregistered test is a one-sided paired permutation test (10 country pairs, exact 2^{10}) of $W_1(\text{generic}) - W_1(\text{region}) > 0$ at $\alpha = 0.05$. Note that with 10 pairs the exact permutation distribution has granularity $1/1024$, so the one-sided p is coarse by construction. **H1**—region-conditioned calibration yields smaller distributional distance than generic calibration—is supported iff this test rejects in the predicted direction; otherwise it stands rejected. Secondary metrics, each reported with 95% bootstrap confidence intervals, are (a) within-country dispersion preservation; (b) between-country heterogeneity, as the Spearman rank correlation of country means; and (c) the EV-alignment gap. Every metric is gated on the output-completeness check (all 1,392 expected generations present).

After the main run we found and fixed two analysis bugs (the script had never been exercised end-to-end): a country filter that emptied the pooled generic arm, and a replicate-grouping error on the generic arm's 8-persona × 10-replicate structure. Independent review then surfaced a deeper problem no analysis fix could cure. The generic arm had reused the same 8 personas identically across all 10 countries, so its per-country W_1 estimates shared one simulated distribution by construction—correlating the paired differences and leaving the generic arm with zero between-country variation by design. Since this is a data-structure defect, we re-ran the generic arm with 8 fresh independent draws from the pooled mixture *per country* (80

Table 1: Per-country Wasserstein-1 distance of simulated to prospect-theory-imputed choice distributions, by arm. Lower is better. n_R / n_G = complete region / generic individuals after parse failures (generic arm: corrected per-country independent draws).

Country	Region	Generic	Uncalibrated	n_R	n_G
Iceland	0.292	0.181	0.442	4	3
Greece	0.312	0.451	0.421	4	6
Brazil	0.333	0.222	0.400	4	3
Hungary	0.200	0.167	0.400	5	6
Pakistan	0.132	0.354	0.254	3	4
Australia	0.346	0.229	0.254	5	6
Germany	0.174	0.438	0.138	3	2
Japan	0.667	0.357	0.200	1	7
Denmark	0.250	0.200	0.200	2	5
Italy	0.333	0.250	0.233	3	6
Mean	0.304	0.285	0.294		

individuals total; `code/run_generic_v2.py`)—the reading of the preregistration's "drawn from the POOLED 27-country distribution" that gives each country an independent generic estimate at the same n as the region arm. Everything else (elicitation, tasks, model, sampling parameters) is identical; only the persona-draw keys and generation seeds are per-country. All generic-arm results below use this corrected data, while the original shared-persona records are retained unmodified as an audit trail. The preregistered hypotheses, metrics, and decision rule are unchanged, and a one-command reproduction package (code, data, verifier; `bash reproduce.sh`) accompanies the paper so every number below can be re-verified independently.

Results

The corrected dataset contains all 1,392 expected generations (48/48 baseline, 480/480 region + 480/480 corrected generic, 384/384 robustness), with parse rates (first Yes/No token found) of 93.8% baseline, 88.3% region, and 91.2% corrected generic. The no-label audit comes back clean: 0 hits across prompt templates, all prompts, and all 1,392 generations.

Primary test: H1 rejected

The three arms land almost on top of each other: mean W_1 0.304 for region, 0.285 for generic, 0.294 for uncalibrated. The paired difference (generic − region) is −0.019, with region favored in only 3 of 10 countries; the one-sided exact permutation test returns $p = 0.631$, far from $\alpha = 0.05$. **H1 is rejected.** We report this as a null, plainly: no detectable region advantage, and the uncalibrated arm matches about as well as either calibrated arm—on this metric, calibration adds nothing over no persona at all. An exploratory sensitivity check that drops the three thinnest region cells (Japan, Denmark, Pakistan) agrees (diff −0.008, $p = 0.539$ on 7 countries).

Secondary results

Robustness. Across Germany and Italy, none of the four elicitation variants gives the region variant a consistent edge over the corrected generic: paraphrase +0.014, seed −0.014, temperature 0.3 +0.042, loss-aversion-only ablation +0.205 (mean W_1 differences, generic − region-variant). Only the ablation shows a sizable gap—on two countries, so suggestive rather than confirmatory.

EV-alignment. Mean share of "yes" answers comes out as 0.62 for region, 0.48 for generic, 0.27 for uncalibrated, 0.50 for prospect-theory-imputed, and 0.67 for EV-optimal. The region arm answers at roughly the EV-optimal rate, the corrected generic arm near the human-predicted rate, and the uncalibrated arm is strongly yes-averse. That runs counter to the preregistered baseline expectation—the classic rationality gap, with the uncalibrated model more EV-rational than humans. We do *not* read this as a non-replication of Liu et al. (2025). Their finding rests on actual human choices (choices13k) on lottery tasks; ours compares a raw-completion Yes/No format against a synthetic CPT baseline. The flip is a cautionary observation about format sensitivity, not evidence against the published gap.

Dispersion and heterogeneity. Within-country dispersion preservation carries no information here: several country cells have near-degenerate simulated distributions, which blows up the log-ratio (region 8.2, generic 12.0, with enormous bootstrap CIs). Between-country heterogeneity is null in both arms—Spearman $\rho = 0.09$ ($p = 0.81$) for region, $\rho = 0.18$ ($p = 0.61$) for generic—with no detectable tracking of real between-country differences. (Thanks to the corrected per-country generic draws this comparison is now fair; in the original shared-persona implementation it was degenerate by construction.)

Discussion

The preregistered bet fails its own test: region-calibrated personas do not significantly improve fit to country-level choice distributions, and the uncalibrated baseline matches about as well as either calibrated arm. The null has three layers, and each teaches something different.

First, the null itself is clean. Under the corrected generic arm there is no directional hint of a region advantage: paired difference −0.019 (3 of 10 countries favor region), $p = 0.631$, and the exploratory thin-cell sensitivity agrees. An earlier analysis had shown a suggestive +0.072 ($p = 0.090$), but that signal was partly an artifact of the shared-persona structure—with independent per-country generic draws, it vanishes. A null, not a near-miss, and we claim nothing more.

Second—and more damaging—is the *uncalibrated* half of the null. Genuine calibration signal should separate the region arm from the no-persona baseline; instead the two tie (0.304 vs. 0.294). Whatever the personas contribute gets

swamped by the model's decisive answering style: simulated per-individual distributions come out extreme-heavy (many 0.0 or 1.0 values), so the model delivers verdicts rather than graded propensities. On this metric, in this format, the persona is nearly inert.

Third, a caution about the rationality gap. The literature's reliable finding—models more EV-rational than humans (Liu et al., 2025)—is measured against actual human choices. Our comparison runs against a synthetic CPT baseline in a raw-completion Yes/No format, so the flipped pattern (uncalibrated yes-averse at 0.27 vs. predicted 0.50) is an apples-to-oranges contrast. We report it as evidence that response format shapes what "human-like" even means—a complication for any calibration agenda built on the gap—rather than as a challenge to the published finding.

For pluralistic alignment, what we have is a boundary condition, not a verdict: region-specific calibration *as implemented here*—small model, qualitative personas, willingness format—does not buy country-level fit. Concretely, Amendment 2 collapsed continuous (λ, ϕ) draws into three fixed qualitative sentences via pooled tertiles, so the region and generic arms differ only in mixture weights over identical sentences. The null is therefore joint over the hypothesis *and* this coarse transmission channel—a finer-grained persona could in principle carry distributional signal that three sentences cannot. This does not show that no regional calibration could work. It shows that the cheapest version does not, and that claims about "culturally calibrated" simulators need preregistered tests with this structure before they are believed.

Concurrent work disclosure. Independently, Yee & Koh (2026) develop profile-based calibration of behavioral parameters (loss aversion, herding, extrapolation, anchoring) in LLMs against generic human benchmarks, documenting the same baseline rationality bias this literature reports; their work appeared on arXiv in February 2026, before this paper's experiments were run (October 2026). Where we differ is in testing *region-specific* calibration to measured cross-country distributions—questions Yee & Koh (2026) do not address. A second concurrent audit reaches a compatible verdict at much larger scale. Chen et al. (2026) construct one-to-one "digital twin" personas for 67,201 Global Preference Survey respondents across 74 countries and find LLM-simulated responses explaining only about 1.3% of individual-level and 4.2% of country-level variance in human economic preferences; in ablations, removing the country label from the persona barely changes alignment, while removing demographic cues degrades it substantially. That models condition on explicit persona content rather than on latent country associations is consistent both with our no-label design and with our result that region-calibrated behavioral profiles add little over an uncalibrated baseline.

Limitations

Several limits deserve plain statement. The most serious is cell thinning: parse failures (11.7% on the corrected main run) left region cells with 1–5 complete individuals per country (Japan: $n = 1$), and a W_1 distance computed from a single draw is not a distribution estimate—so the primary test's inputs are degraded at exactly the noisiest cells, which the paired test accounts for only partially. Dropping unparseable generations was a forced, non-preregistered choice, and the missingness is non-random on top of that: failures concentrate on gain tasks (up to 29% on the $65 gain task vs. near-zero on loss tasks) and on the middle risk persona ("somewhat cautious" parses at 49–65% vs. 86–100% for the extreme levels). Cell thinning is therefore not ignorable—though the exploratory thin-cell sensitivity (dropping Japan, Denmark, Pakistan) agrees with the main result. A second caveat concerns the generic-arm correction: while it implements the preregistration's per-individual pooled draws, it required new data collection after the original results were seen. It is a reviewer-mandated repair rather than a preregistered step, and a fully preregistered replication would re-run both arms. Third, Qwen3-1.7B is a small model on CPU, and larger models may show different calibration dynamics. Fourth, the Yes/No willingness format (Amendment 2) elicits more coarsely than the preregistered lottery-choice format; format effects are visible in the flipped rationality-gap pattern. Fifth, the "real" choice distributions are deterministic prospect-theory predictions from reconstructed—not observed—within-country parameter distributions, twice removed from actual human choices. The within-SD sensitivity check moves the paired difference only from -0.018 to -0.032 (ordering unchanged), but absolute fit levels remain fragile to the reconstruction. The INTRA samples, sixth, are economics students rather than representative populations, and within-country distributions are reconstructed from published standard errors rather than observed microdata. Finally, the exact probability-weighting form in Rieger et al. (2017) is not public; the Tversky–Kahneman form used here is a disclosed assumption.

Conclusion

We preregistered a clean test of a natural hypothesis—that calibrating LLM personas to region-specific measured preference distributions reproduces country-level choice heterogeneity better than a generic-human calibration. Run on Qwen3-1.7B, the test returned a null: with the corrected per-country generic arm, the paired difference is -0.019 ($p = 0.631$)—no detectable region advantage, and calibration adds nothing over an uncalibrated baseline (0.304 vs. 0.285 vs. 0.294). The null's value is discipline. It tells the pluralistic-alignment agenda what the cheapest version of regional calibration does *not* buy, under exactly the conditions a preregistration makes checkable. If culturally calibrated simulators are to be more than a slogan, they will need to pass tests built with this structure.

Declaration of generative AI use

The author used the AI tool Muse (Muse Spark 1.3, Meta) for drafting, grammatical editing, code validation, and literature search. The contents are reviewed, edited, and attested for accuracy by the author.

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